

ECON 101—Introduction to Microeconomics
Fall 2024
Midterm I

September 19, 2024

Name:

NU ID number:

Instructions:

- Answer all questions. **All answers should be written in the answer booklet.** You can use as many scratch paper for your preliminary calculations, but that work will not be considered. I will only consider the answers written in the exam booklet.
- You are not allowed to unstaple the exam booklet or the answer booklet.
- The exam has a duration of 60 minutes.
- You can request a bathroom break, but only one person can be out at any given time.
- On your desk, you are allowed to have your ID, writing instruments, a ruler, and a simple calculator (no graphing calculators). You may also have a simple watch if needed (smartwatches are not allowed).
- Please place all other personal items, including smart devices, to the side of the room.
- If you complete the exam early, you may leave the room. but only within the first 45 minutes. Otherwise, please remain in the room quietly until the exam concludes.
- Crying is allowed, but please do so quietly to avoid disturbing others.
- Do not use the exam paper to wipe away tears.
- Good luck and may the elasticity be with you!

Part I: Multiple Choice

Please record all answers in the front sheet. Each question is worth 2 points for a total of 12 points.

1. If an individual lives longer than expected, having spent their savings, and thus requiring financial assistance from their children, then the opportunity cost of the resources they receive from their children is _____. If the individual dies earlier than expected, leaving an inheritance to their children, the corresponding opportunity cost of the resources inherited by their children is _____.
 - (a) Not having died when expected. The greater consumption that the children will now be able to enjoy.
 - (b) The excess of their own consumption that led them to financially depend on their children. The poorly used savings.
 - (c) The interest that they stopped earning because their savings were depleted prematurely. The cost of the funeral.
 - (d) At least equal to the unrealized consumption by their children. At least equal to the consumption they did not realize.

2. In the town of NUtopia, the townspeople like to consume lattes. Suppose there are two types of consumers: regular coffee drinkers (regulars R), and coffee snobs (snobs S). Suppose that the market equilibrium happens at the same price and quantity for both the regular and snob's market. at the *market equilibrium*, the price elasticity of demand for regular drinkers is $\epsilon_{D,R}^P = -2$ and the price elasticity of demand for snobs is $\epsilon_{D,S}^P = -1.2$. If the supply of lattes in NUtopia is perfectly elastic and if the government of NUtopia implements a subsidy of \$4 in this market we can be sure that: The increase in the consumer surplus for coffee snobs will be _____ than the increase in the consumer surplus for regulars. The dead-weight loss will be greater in the market of _____.
 - (a) lower than; regulars.
 - (b) greater than; regulars.
 - (c) greater than; snobs.
 - (d) lower than; snobs.

3. An NU Economics professor developed gastritis after years of hearing his students say, "Economics is not applicable to real life." His doctor prescribed Omeprazole (Омепразол). He noticed that due to the tax imposed by the government, Omeprazole has become more expensive. Nevertheless, he continues to buy it. We can infer that:
 - (a) Omeprazole, by not having many substitutes, has an elastic supply curve.
 - (b) The professor has an inelastic demand for Omeprazole.
 - (c) The professor has an elastic demand for Omeprazole.
 - (d) Omeprazole, by not having many substitutes, has an inelastic supply curve.

Read carefully the following text and answer questions 4–5.

In Seville, Figaro, the versatile barber, and Dr. Bartolo, the wealthy but conniving guardian of Rosina, are both involved in two key activities: cutting hair and writing love letters. Figaro is excellent at cutting hair and is reasonably good at writing persuasive love letters. Dr. Bartolo, on the other hand, is not great

at either task, but he is slightly better at writing letters than cutting hair. In one day:

- Figaro can cut 6 heads of hair (H) or write 4 love letters (L).
- Dr. Bartolo can cut 2 heads of hair (H) or write 2 love letters (L).

In the questions below, put H in the horizontal axis and L in the vertical axis.

Answer the following **two** questions, 4–5

4. The production possibilities frontier of Figaro is represented by the equation _____, the production possibilities frontier of Dr. Bartolo is represented by the equation _____. For Figaro the point $(H, L) = (3, 2)$ is _____, for Dr. Bartolo the point $(H, L) = (1, 1)$ is _____.

- (a) $L = -\frac{2}{3}H + 4$; $L = 2 - H$; feasible and efficient; feasible and efficient.
- (b) $H = 6 - \frac{2}{3}L$; $H = 2 - L$; not feasible; feasible and efficient.
- (c) $L = -\frac{3}{2}H + 4$; $L = 2 - H$; not feasible; feasible and efficient.
- (d) $H = 6 - \frac{3}{2}L$; $H = 2 - L$; feasible and efficient; feasible and inefficient

5. Does anyone of them has a comparative advantage? _____. Does any one of them has an absolute advantage? _____. If each of them specializes in the production of the good/service in which they have a comparative advantage, and they decide to trade between themselves, then Figaro would specialize in _____ and Dr. Bartolo in _____, and the possible terms of trade (opportunity cost of H in terms of L when trading) would be _____

- (a) Yes; yes; cutting hair; love letters, $\frac{5}{6}$.
- (b) There are no comparative advantages.
- (c) Yes; no; cutting hair; love letters; 1.
- (d) Yes; yes; love letters; cutting hair, $\frac{3}{4}$,

6. Medina told Akmaral that since the cinema ticket price increased from 1,450 to 2,900 tenge, she now goes only half as often as she used to. With this information Akmaral knows that:

- (a) The price elasticity of demand of Medina cannot be calculated with the data we have.
- (b) The price elasticity of demand of Medina, in absolute value, is $\frac{1}{2}$.
- (c) Medina is not a true fan of cinema.
- (d) The price elasticity of demand of Medina, in absolute value, is 2.

End of Multiple Choice Section: Please don't forget to record your answers in the answer sheet!

Continue to Part II

Part II: Long Problems

Please answer *only* in the answers booklet. Be neat and give complete arguments. Points for each item are indicated. You can use as much scratch paper as you need but your scratch work will not be considered.

The whimsical land of Ekonomika

In the land of **Ekonomika**, the fair, wise, and highly knowledgeable in Economics, Princess Medina rules over her subjects. The law in this land is maintained, harshly but fairly, by Captain Akmaral and General Nazima (i.e., cheating is illegal and any suspected cheating is dealt with the harshest penalties).

In Ekonomika, unicorns are not just mythical creatures—they are hard working entrepreneurs who produce *Sparkle Dust*, which is a crucial ingredient for wizard's potions and the lattes that the Gen Z and Gen α citizens of Ekonomika like so much, as well as having many other uses. Suppose that the market for Sparkle Dust can be modeled using the tools that you have learned in ECON101.

Lord Treasurer Bagdat claims that the market for Sparkle Dust can be modeled according to the following supply and demand equations:

$$\begin{aligned}Q_d^0 &= 240 - 2P \\ Q_s^0 &= 4P - 80\end{aligned}$$

Recently, the unicorns discovered that listening to Taylor Swift songs while working doubles their Sparkle Dust production. However, constant exposure to Taylor Swift songs causes the unicorns to lose their magical rainbow colors, turning them plaid instead. This has two effects:

- (Event 1) Listening to Taylor Swift doubles unicorn productivity:

$$Q_s^1 = 8P - 160.$$

- (Event 2) Gen α consumers now prefer *Plaid Unicorn Sparkle Dust*, believing it to be more authentic, i.e., they claim that it is *mad Sigma*. This shifts the demand for Sparkle Dust:

$$Q_d^1 = 320 - 2P.$$

However, there is a problem! The Council of Wizards, led by Archmage Nuray, is concerned that the overproduction of Sparkle Dust will have deleterious effects on the space-time continuum. (Event 3) Archmage Nuray recommends to put a *per-unit tax* of \$10 on Sparkle Dust to curb its production.

(Event 4) Additionally, the evil wizard Sultan spreads a rumor that consuming Sparkle Dust grants immortality, which further affects the demand to:

$$Q_d^2 = 400 - 2P.$$

After all this craziness, Lord Treasurer Bagdat gathers two experts on the matter, Minister Abuzer and Exchequier Rakhim, to analyze this mess. In this problem, you will help them understand what is going on in the market for Sparkle Dust in the land of Ekonomika! In each task below assumes that the events prior to the one being considered remain.

1. (5 points) Calculate the **initial equilibrium** before any changes occur.
2. (5 points) Determine the new equilibrium price and quantity after the productivity boost (Event 1).
3. (5 points) Find the equilibrium price and quantity after the Gen α demand increase (Event 2).
4. (5 points) Calculate the equilibrium price and quantity after the per-unit tax is imposed (Event 3).
5. (5 points) Calculate the final equilibrium price and quantity after the immortality rumor affects demand (Event 4.)

6. (10 points) Compute consumer surplus, producer surplus, and the dead-weight loss, if any, at each stage.
7. (8 points) Discuss the welfare implications for the inhabitants of Ekonomika. In your argument, you must use the concepts of tax incidence, and market efficiency.

Note: Show all your calculations and illustrate each stage with supply and demand graphs.

The time-traveling labor market of South Park

In the quaint town of South Park, the demand for snow-shoveling services is at an all-time high during winter. The demand for labor is fixed at $Q = 100$ jobs per day, regardless of the wage rate.

Initially, only the local workers, affectionately known as **Townies**, supply labor in this market. Their supply curve is given by:

$$S_{\text{Townies}} : W = 0.5Q_T + 5$$

where: W is the wage rate in dollars per hour, and Q_T is the quantity of labor supplied by the Townies.

One frosty morning, a group of time-traveling workers from the future, dubbed **Future Folks**, arrive and begin offering their labor. Due to difficulties in obtaining formal snow-shoveling jobs, they can supply a maximum of $Q_F^{\max} = 30$ jobs per day. Their supply curve is given by

$$S_{\text{Future Folks}} : W = 0.3Q_F + 2 \quad \text{with } Q_F \leq 30.$$

where: Q_F is the quantity of labor supplied by the Future Folks.

The Townies are up in arms, exclaiming, "They took our jobs!"

To address the uproar, the South Park town council decides to implement a minimum wage policy set at $W_{\min} = \$50$ per hour.

After this policy is enacted, the townies are still up in arms claiming: **"They took our jobs!"**



1. (20 points) **Without** the minimum wage, determine the equilibrium wage rate and the total quantity of labor supplied. How many jobs do the Townies and Future Folks each supply at this wage?
2. (25 points) **With** the minimum wage of \$50 per hour in place, analyze its effect on the labor market. Consider the following allocation methods:
 - (a) Method 1: Assuming no hiring bias, jobs could be allocated proportionally on a first-come, first-served basis. This means that given the total labor supply at the minimum wage, you will compute the proportion of townies and future-folk that want to offer their labor. You will use this proportion to compute how many townies and future-folk get the 100 available jobs.
 - (b) Method 2: Given the local uproar, employers prioritize Townies.

Discuss, under each method, whether the Townies' complaint that "They took our jobs!" is justified. Does the minimum wage help or hurt their employment prospects?

Note: Show all your calculations and illustrate each stage with supply and demand graphs.